

SAMSUN, Turkey

WMO: 170300

Lat: 41.3439N

Lon: 36.2562E

Elev: 4

StdP: 101.28

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	0.9	-8.6	1.8	5.2	-6.7	2.1	5.7	9.2	10.3	8.2	11.2	2.4	290	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	29.2	22.5	28.4	22.3	27.7	22.0	24.0	27.8	23.5	27.3	22.9	26.8	2.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.7	17.4	26.9	22.1	16.8	26.6	21.4	16.1	26.3	72.3	28.0	70.0	27.4	68.0	26.8	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.6	5.5	4.7	-1.6	33.5	2.0	1.7	-3.0	34.7	-4.2	35.6	-5.3	36.6	-6.8	37.8	
			DB												
			WB	-3.3	25.4	2.2	0.9	-4.9	26.0	-6.1	26.6	-7.3	27.1	-8.9	27.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	7.8	8.2	9.1	12.0	16.4	21.0	23.6	24.4	20.9	16.8	12.9	9.6
	DBStd	6.65	3.68	4.16	3.67	3.47	2.85	2.34	1.55	1.78	2.56	3.03	3.69	3.84
	HDD10.0	312	89	80	62	15	0	0	0	0	0	0	13	53
	HDD18.3	1684	327	285	286	194	73	5	0	7	68	169	271	
	CDD10.0	2232	21	28	36	74	200	329	421	446	327	210	99	42
	CDD18.3	561	0	1	1	3	14	84	162	187	84	19	4	1
	CDH23.3	3247	0	1	7	32	38	315	1036	1379	369	52	18	0
	CDH26.7	443	0	0	1	11	7	27	116	239	31	11	1	0
Wind	WSAvg	1.8	2.2	2.0	1.7	1.4	1.3	1.6	2.0	1.9	1.7	1.5	1.7	2.1
Precipitation	PrecAvg	703	67	52	62	58	51	46	31	37	51	81	83	81
	PrecMax	1012	154	126	142	146	188	151	116	164	133	257	350	238
	PrecMin	503	4	14	5	7	8	3	0	0	4	4	5	19
	PrecStd	109	33	25	28	30	33	28	26	37	33	53	55	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	22.3	23.9	27.2	27.2	28.8	29.6	31.0	29.2	27.8	25.6	20.9
		MCWB	10.6	11.6	12.9	15.5	18.4	22.2	22.8	23.2	20.6	17.7	13.6	11.8
	2%	DB	17.3	19.5	19.9	22.5	23.7	27.2	28.7	29.5	27.5	24.4	22.3	18.9
		MCWB	9.3	10.7	11.5	14.7	17.8	21.9	22.6	23.1	21.0	18.2	13.0	11.0
	5%	DB	15.2	17.0	17.2	19.2	22.1	26.2	27.9	28.7	26.4	22.8	20.0	17.1
		MCWB	8.4	9.6	10.7	13.7	17.4	21.3	22.4	22.8	20.9	18.0	12.8	10.4
	10%	DB	13.3	14.7	14.5	17.0	20.7	25.1	27.2	28.0	25.5	21.3	18.3	15.2
		MCWB	7.8	8.5	9.7	12.5	16.9	20.4	22.0	22.5	20.6	17.3	12.3	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	12.5	13.5	16.6	19.9	23.9	24.6	25.2	23.2	21.6	16.5	13.0
		MCDB	16.7	20.0	21.7	23.9	23.8	27.4	27.5	28.8	26.5	24.2	19.7	18.3
	2%	WB	10.2	11.0	12.1	15.2	18.7	22.9	23.8	24.2	22.5	19.9	15.1	12.0
		MCDB	14.9	18.2	18.9	21.6	22.3	26.3	27.4	28.2	26.1	22.8	18.8	16.7
	5%	WB	9.3	10.0	11.0	13.9	17.9	21.9	23.3	23.7	21.9	18.7	14.1	11.1
		MCDB	13.2	15.5	16.0	18.4	21.1	25.7	27.0	27.7	25.6	21.6	18.1	15.2
	10%	WB	8.5	9.1	10.2	13.0	17.1	21.0	22.6	23.2	21.1	17.8	13.2	10.3
		MCDB	12.0	13.5	13.7	16.5	20.3	24.7	26.6	27.2	25.1	20.7	17.4	14.3
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.1	6.6	6.4	5.9	6.2	6.5	6.5	6.8	6.1	6.8	5.7
		MCDBR	8.8	10.6	12.0	11.2	7.9	7.3	7.0	6.8	7.9	8.5	9.9	8.5
		MCWBR	4.3	4.9	5.5	5.4	4.4	5.3	5.1	4.9	5.8	5.1	5.1	4.2
	5% WB	MCDBR	7.2	9.2	10.4	9.4	6.7	6.9	6.6	6.7	7.2	6.8	7.7	7.2
		MCWBR	4.1	4.7	5.4	5.2	4.3	5.3	5.3	5.2	6.0	5.0	5.6	4.1
Clear-Sky Solar Irradiance	taub	0.355	0.391	0.436	0.480	0.450	0.423	0.439	0.449	0.419	0.409	0.368	0.354	
	taud	2.321	2.214	2.113	2.032	2.202	2.326	2.302	2.264	2.337	2.303	2.365	2.347	
	Ebn at Noon	777	799	803	795	832	854	836	815	816	772	757	743	
	Edh at Noon	89	115	142	165	142	126	128	129	112	104	85	81	
All-Sky Solar Radiation	RadAvg	1.59	2.33	3.38	4.70	5.73	6.60	6.58	5.77	4.43	2.92	2.03	1.42	
	RadStd	0.17	0.28	0.26	0.48	0.57	0.52	0.34	0.33	0.31	0.27	0.23	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	-82	-143	+106	+66

Nomenclature: See separate page